

CAMPUS SYNC

Verified Co-living & Accommodation Platform for Employees

1. Business Requirement

Employees and candidates relocating to a new city for work need safe, trustworthy, and conveniently located shared accommodation (PG / co-living) near their assigned office. Public rental platforms are open to everyone, list unverified rooms, and give no assurance that a flatmate is a genuine colleague.

Campus Sync addresses this with a closed, verified platform — exclusive to the organization's employees and candidates — where they can both offer and discover shared rooms with confidence.

The Problem :

Every year, Cognizant freshers receive offer letters and are assigned to offices in cities they have never lived in. Within days they face urgent questions:

- Where do I stay? Online PG listings from strangers are risky and unverified.
- Who do I stay with? Living with a verified colleague is far safer than a random tenant.
- How do I commute? Shuttle timings, auto routes, and travel costs near the office are unknown until you actually arrive.
- Is this PG good? Public reviews on Google Maps are from strangers — not from Cognizant employees who have actually lived there.

Most freshers spend their first 2–4 weeks in expensive hotels, make poor accommodation decisions, and feel isolated because they have no peer connections before Day 1. The information they need already exists — it lives in the experience of current employees. There is simply no secure channel to access it.

2. Proposed Solution

Campus Sync is a verified, closed web platform connecting Providers (a vacant bed) with Seekers (need a room). Access is gated by ID verification, so every user is a genuine colleague.

It is a gated peer-to-peer platform where identity is verified at the door. Every user — whether a current employee or a new joiner — must prove their Cognizant identity before accessing any feature. Current employees share what they know. New joiners access it securely. Because every contribution is tied to a verified Cognizant ID, the trust is built-in.

- **Verified onboarding** — sign-up completes only if the ID exists in the Super Admin DB and is unused.
- **Server-side gender matching** — the backend returns only matching or co-ed listings; ineligible ones never reach the client.
- **Location-aware discovery** — seekers filter by office location for commutable rooms.

- **Mandatory quality ratings** — a review and 1–5 star food/service ratings are required before a listing goes live.
- **Trusted forum + auto-cleanup** — verified peers exchange advice; a scheduled job removes unconfirmed stale posts.

2.1 Goals & Roles

- Restrict access to verified employees/candidates via ID verification against the Super Admin's records.
- Let users with a vacant bed publish listings (Provider) and seekers discover relevant ones (Seeker).
- Auto-match seekers to gender-appropriate, location-relevant listings; mandate food & service ratings on every post.
- Provide a trusted, categorized community forum and keep the feed fresh via automated cleanup of stale posts.

Role	Responsibility
Super Admin	Owns the secure database of valid Employee/Candidate IDs used for verification.
Provider	Lists an available bed with details, accommodation type, and mandatory ratings.
Seeker	Browses, filters, and contacts providers to finalize a room offline.
System (Scheduler)	Background job that prompts and removes stale listings.

2.2 Functional Requirements

ID	Requirement
FR-1	Validate Employee/Candidate ID against the Super Admin DB; allow only valid, unused IDs.
FR-2	Capture profile (name, email, phone, password, gender) and authenticate on login.
FR-3	Browse listings by sharing type (Double / Triple).
FR-4	Server-side gender filtering — users see only matching or co-ed listings.
FR-5	Filter listings by selected office location.
FR-6	Create a listing with mandatory food/service ratings and review.
FR-7	View provider phone number to coordinate offline.
FR-8	Post and reply to categorized questions in the Doubts forum.
FR-9	Prompt providers at 12 days; auto-remove unconfirmed listings after a 2-day grace period.

3. DATABASE SCHEMA

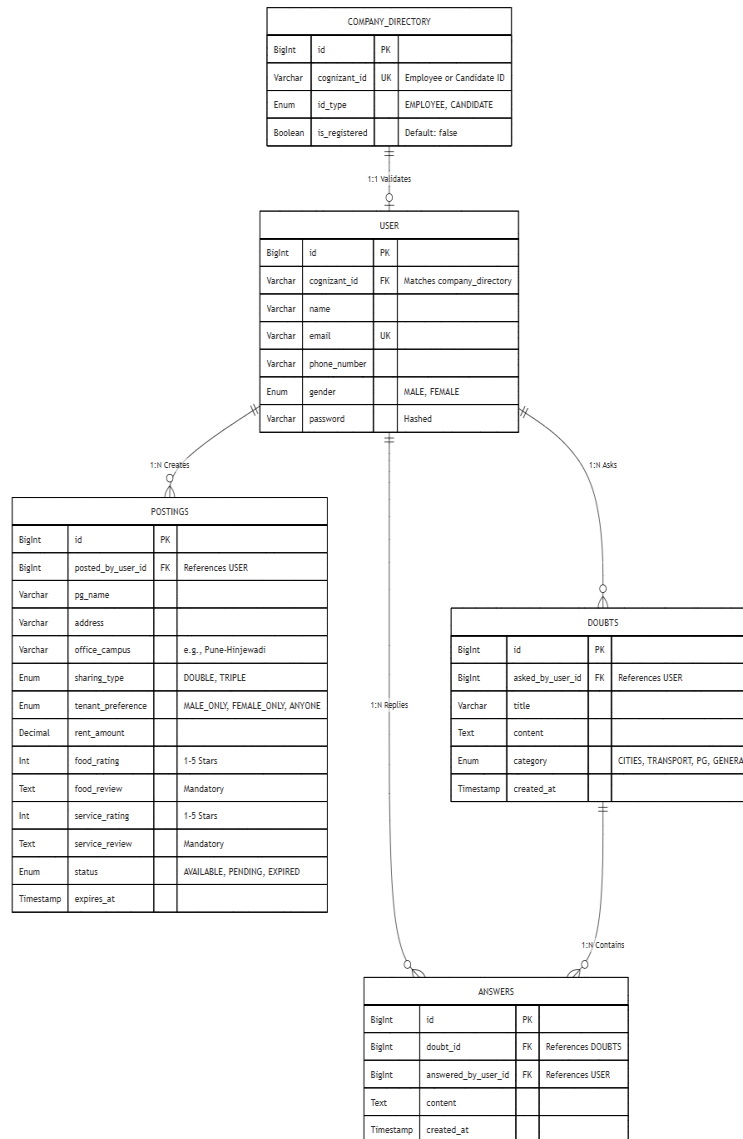


Table Name	Fields (Columns)
COMPANY_DIRECTORY	id, cognizant_id, id_type, is_registered
USER	id, cognizant_id, name, email, phone_number, gender, password
POSTINGS	id, posted_by_user_id, pg_name, locality_and_landmark, office_campus, sharing_type, tenant_preference, rent_amount, food_rating, food_review, service_rating, service_review, image_url, status, expires_at
DOUBTS	id, asked_by_user_id, title, content, category, created_at
ANSWERS	id, doubt_id, answered_by_user_id, content, created_at

Technical Design Document

1. Technology Stack

A classic three-tier web architecture: an Angular single-page client, a stateless Spring Boot REST service layer, and a MySQL database.

Layer	Technology	Role
Frontend	Angular (TypeScript)	SPA for sign-up, login, dashboard, listings, and forum; consumes REST over HTTPS.
Backend	Spring Boot — Spring REST	JSON REST endpoints, routing, validation, and business logic.
Persistence	Spring Data JPA (Hibernate)	ORM mapping of entities to MySQL; repository abstraction.
Database	MySQL	Relational store for users, IDs, listings, and forum data.
Security	Spring Security + JWT	Stateless token auth; BCrypt password hashing.
Scheduling	Spring @Scheduled	Background job for automated listing cleanup.

2. Key Workflows

1. Onboarding

1. User enters Employee/Candidate ID; backend checks `valid_ids` (exists and unused).
2. User submits profile; backend hashes the password, creates the user, marks the ID used.
3. User logs in, gets authenticated and lands on the dashboard.

2. Browsing (Seeker)

4. User selects Double/Triple sharing; backend applies the gender filter from the profile.
5. User selects office location; cards show rent, amenities, and mandatory ratings.
6. User opens a card, gets the provider's phone number, and finalizes offline.

3. Listing (Provider)

7. User enters PG details, rent, sharing type, and accommodation type (Male/Female/Anyone).
8. System enforces a review and 1–5 star food/service ratings before submit; the post goes live.

4. Forum & Auto-Cleanup

9. Users post/answer categorized questions (Transport/General) in the Doubts tab.
10. A daily prompt to expire any posting if expriy date reached.